Self-scaled bound of the regular dodecahedron

We provide an exact certificate for the self-scaled bound of the slack matrix X of the regular dodecahedron. We use the matrix `Sdodecahedron` in `SlackMat.mat`, with the row and column ordering of the accompanying repository. Its exact entries belong to $\{0, 3-\sqrt{5}, \sqrt{5}-1, 2\}$, each occurring 60 times. Put $\phi = (1+\sqrt{5})/2$ and $b = 3-\sqrt{5} = 2\phi^{-2}$. The positive entries are $b, b\phi, b\phi^2$.

Recall that

$$\mathrm{SSB}(X) = \max_{L}\{\langle L, X\rangle : \langle L, R\rangle \le 1 \text{ for all } R \in \mathcal{R}(X)\}, \qquad \mathcal{R}(X) = \{uv^\top : u, v \ge 0,\ uv^\top \le X\}.$$

The script `CertificateDodecahedron.py` checks the exact geometry, symmetries, finite enumeration and decomposition below using integer arithmetic.

Proposition 1. *The slack matrix X satisfies* $\mathrm{SSB}(X) = 42/5$. *An optimal dual certificate is*

$$L^*_{ij} = \begin{cases} 1/(10b) = \phi^2/20 & \text{if } X_{ij} = b, \\ 1/50 & \text{if } X_{ij} = 2, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. The objective value of L^* is

$$\langle L^*, X\rangle = \frac{60b}{10b} + \frac{60\cdot 2}{50} = 6 + \frac{12}{5} = \frac{42}{5}.$$

We first verify its feasibility. For a nonzero admissible matrix, normalize its factors so that $\max_i u_i = 1$. Then $0 \le u_i \le 1$ and $0 \le v_j \le 2$. Since $L^* \ge 0$, we may increase each positive v_j to at least b, and then each positive u_i to at least $b/2 = \phi^{-2}$, without decreasing the objective. These changes preserve admissibility: on the positive supports, every entry of X is at least b, while a changed product is at most b.

On the positive supports, write $u_i = \phi^{-s_i}$ and $v_j = b\phi^{t_j}$, where $s_i, t_j \in [0,2]$. If $X_{ij} = b\phi^{k_{ij}}$, admissibility is equivalent to $t_j - s_i \le k_{ij}$, with $k_{ij} \in \{0,1,2\}$. For $\theta \in [0,1)$, define

$$s_i(\theta) = \lfloor s_i + \theta \rfloor, \qquad t_j(\theta) = \lfloor t_j + \theta \rfloor.$$

These integer vectors remain in $[0,2]$ and preserve every difference inequality, since its right-hand side is an integer. Their average is (s,t). Moreover, the function

$$f(s,t) = b \sum_{i\in\mathrm{supp}(u),\, j\in\mathrm{supp}(v)} L^*_{ij}\phi^{t_j - s_i}$$

is convex. Jensen's inequality therefore gives $f(s,t) \le \int_0^1 f(s(\theta), t(\theta))\, d\theta$. An index with $u_i = 1$ has $s_i = 0$ and remains fixed. Thus it suffices to consider

$$u_i \in \{0, \phi^{-2}, \phi^{-1}, 1\}, \qquad v_j \in \{0, b, b\phi, b\phi^2\}.$$

The script constructs a group G of 120 simultaneous row and column permutations preserving X, and checks that its action on the rows is transitive. These permutations also preserve L^*. We may therefore assume $u_1 = 1$. Row 1 contains three entries of each of the four values. The corresponding choices for v_j have respectively $1, 2, 3, 4$ elements, which gives $2^3 3^3 4^3 = 13\,824$ configurations.

For each configuration, choose every u_i as its largest feasible value in $[0, 1]$. If a positive v_j meets a zero entry of row i, set $u_i = 0$. Otherwise, writing $v_j = b\phi^{t_j}$ on its positive support, take

$$s_i = \max\{0, \max_{j : v_j > 0} (t_j - k_{ij})\}, \qquad u_i = \phi^{-s_i}.$$

The zero vector v contributes objective zero. All other configurations can be evaluated exactly from

$$50\langle L^*, uv^\top \rangle = 5 \sum_{\substack{u_i v_j > 0 \\ X_{ij} = b}} \phi^{t_j - s_i} + 2 \sum_{\substack{u_i v_j > 0 \\ X_{ij} = 2}} \phi^{t_j - s_i - 2}.$$

Every power occurring here belongs to $\mathbb{Z} + \mathbb{Z}\phi$. The exhaustive enumeration gives a maximum of exactly 50, proving $\langle L^*, uv^\top \rangle \leq 1$ and hence $\mathrm{SSB}(X) \geq 42/5$.

For the upper bound, define $R_A = u_A v_A^\top$ and $R_B = u_B v_B^\top$ by

$$\begin{aligned}
(u_A)_i &= 1 && i \in \{1, 2, 6, 8, 9, 12, 13, 14, 16, 17\}, \\
(v_A)_j &= b && j \in \{3, 8\}, \\
(u_B)_i &= 1 && i \in \{1, 15\}, \\
(u_B)_i &= \phi^{-1} && i \in \{4, 10, 14, 16\}, \\
(u_B)_i &= \phi^{-2} && i \in \{8, 9\}, \\
(v_B)_j &= 2 && j \in \{2, 7\}, \\
(v_B)_j &= b\phi && j \in \{3, 6\},
\end{aligned}$$

with all other entries zero. Indices start at 1. The script verifies $0 \leq R_A, R_B \leq X$ and the exact identity

$$X = \sum_{g \in G} \left(\frac{R_A^g}{50} + \frac{R_B^g}{20} \right).$$

Consequently, every feasible L satisfies

$$\langle L, X \rangle \leq 120 \left(\frac{1}{50} + \frac{1}{20} \right) = \frac{42}{5}.$$

Together with the lower bound, this proves the equality. □

Verification Run `python CertificateDodecahedron.py`. Only NumPy is required. The script contains the exact geometry and two permutation generators, reconstructs the slack matrix and checks both bounds without an optimization solver or an external data file. Run Python without the `-O` option, since the verification uses assertions.